

Quantum Physics Notes

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Contents

1	Basic Concept	2
1.1	Schrodinger Equation	2
2	Probability Current	2
3	Generalized Lagrange Equation	4
4	Spin	6
4.1	Rotation of spin	6
4.2	Expression for the eigenstates of spin in any direction	7
4.3	Zeeman effect	7
4.4	Spin orbital coupling	8
4.5	Larmor precession	8
5	Periodic crystal	9
5.1	Bloch Basis	9
5.2	1D Tight Binding Model	10
6	Scattering	12
7	Transport Property	13
7.1	Boltzmann Transport Equation	13
7.2	Intrinsic Anomalous Hall Effect	13

1 Basic Concept

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1.1 Schrodinger Equation

Here we don't derive the basic Schrodinger equation, only use it as the basic axiom. Assuming the particle can be represented as a complex function, which is $\Psi(\mathbf{r}, t)$. The particle's position is described probabilistically, and the specific probability density at a given point in space $\mathbf{r}_0$ is $\rho(\mathbf{r}_0, t) = \Psi^*(\mathbf{r}_0, t)\Psi(\mathbf{r}_0, t) = |\Psi(\mathbf{r}_0, t)|^2$. The evolution of the particle can be described by time dependent Schrodinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi \quad (1.1)$$

The $\hat{H}$ operator is $\frac{\hat{P}^2}{2m} + \hat{V}$. Assuming the wavefunction can be written in the form as $\Psi = e^{-\frac{iEt}{\hbar}}\psi$, where E is the energy. ψ and $\hat{V}$ are independent on time. This form indicate that, the state only rotates in the complex space because the time only contribute to a change on the phase factor. Then the time dependent Schrodinger equation can be written as the time independent form:

$$\begin{aligned} i\hbar \frac{\partial(e^{-\frac{iEt}{\hbar}}\psi)}{\partial t} &= e^{-\frac{iEt}{\hbar}} \hat{H}\psi \\ e^{-\frac{iEt}{\hbar}} E\psi &= \hat{H}\psi \\ E\psi &= \hat{H}\psi \end{aligned} \quad (1.2)$$

This is the time independent Schrodinger equation.

2 Probability Current

Since the position of the particle can be described by probability distribution, then we can define a quantity, probability current $\mathbf{J}$ to denote that. The continuity equation requires that:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad (2.1)$$

And the probability density is $\psi^*\psi$. Thus, we have:

$$\begin{aligned} -\nabla \cdot \mathbf{J} &= \frac{\partial}{\partial t}(\psi^*\psi) \\ -\nabla \cdot \mathbf{J} &= \psi \frac{\partial}{\partial t}\psi^* + \psi^* \frac{\partial}{\partial t}\psi \end{aligned} \quad (2.2)$$

According to the time dependent Schrodinger equation, when the potential operator $\hat{V}$ is real, we have:

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2 \nabla^2}{2m} + \hat{V}\right)\psi \quad (2.3)$$

$$-i\hbar \frac{\partial \psi^*}{\partial t} = \left(-\frac{\hbar^2 \nabla^2}{2m} + \hat{V}\right)\psi^* \quad (2.4)$$

Substituting back to the continuity equation, we have:

$$\begin{aligned}
-\nabla \cdot \mathbf{J} &= \psi \left[\left(\frac{\hbar \nabla^2}{2mi} - \frac{\hat{V}}{i\hbar} \right) \psi^* \right] + \psi^* \left[\left(-\frac{\hbar \nabla^2}{2mi} + \frac{\hat{V}}{i\hbar} \right) \psi \right] \\
&= \frac{\hbar}{2mi} (\psi \nabla^2 \psi^* - \psi^* \nabla^2 \psi) \\
&= \frac{\hbar}{2mi} \nabla \cdot (\psi \nabla \psi^* - \psi^* \nabla \psi)
\end{aligned} \tag{2.5}$$

Thus we have the expression for the probability current:

$$\mathbf{J} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \tag{2.6}$$

3 Generalized Lagrange Equation

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Starting from the principle of the least action, which has the definition:

$$S[t] = \int_{t_1}^{t_2} L(\mathbf{x}, \dot{\mathbf{x}}, t) dt \quad (3.1)$$

Here the t is the parameter, usually we use it as time. $\mathbf{x}$ is the generalized coordination, and L is the Lagrange. $\mathbf{x}$ is t dependent which means $\mathbf{x}(t)$. According to the principle of the least action, we have:

$$\delta S = 0 \quad (3.2)$$

We only assume the variation of x , which gives:

$$\begin{aligned} \delta S &= \int_{t_1}^{t_2} \delta L(\mathbf{x}, \dot{\mathbf{x}}, t) dt \\ &= \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i} \delta x^i + \frac{\partial L}{\partial \dot{x}^i} \delta \dot{x}^i + \frac{\partial L}{\partial t} \delta t dt \\ &= \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i} \delta x^i + \frac{\partial L}{\partial \dot{x}^i} \delta \dot{x}^i dt \end{aligned} \quad (3.3)$$

And we have:

$$\begin{aligned} \int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}^i} \delta \dot{x}^i dt &= \left[\frac{\partial L}{\partial \dot{x}^i} \delta x^i \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^i} \right) \delta x^i dt \\ &= - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^i} \right) \delta x^i dt \end{aligned} \quad (3.4)$$

And now we have:

$$\delta S = \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i} \delta x^i - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^i} \right) \delta x^i dt = 0 \quad (3.5)$$

Thus we have the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^i} \right) - \frac{\partial L}{\partial x^i} = 0 \quad (3.6)$$

We can define $\frac{\partial L}{\partial \dot{x}^i} = p^i$, applying Legendre transformation, we have Hamiltonia:

$$H(\mathbf{x}, \mathbf{p}) = p_i \dot{x}^i - L(\mathbf{x}, \mathbf{p}) \quad (3.7)$$

Actually the term $p_i \dot{x}^i$ is the so called symplectic potential, and we can replace the $\mathbf{x}$ and $\mathbf{p}$ in a very general form $\boldsymbol{\xi}$, so the Lagrange can be written as:

$$L = -a_i(\boldsymbol{\xi}) \dot{\xi}^i - H(\boldsymbol{\xi}) \quad (3.8)$$

Using this form, we can substitute it into the Euler-Lagrange equation, which has:

$$\frac{\partial L}{\partial \xi^i} = -\frac{\partial a_j}{\partial \xi^i} \dot{\xi}^j - \frac{\partial H}{\partial \xi^i} \quad (3.9)$$

$$\frac{\partial L}{\partial \dot{\xi}^i} = -a_i \quad (3.10)$$

So we have:

$$\begin{aligned}\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\xi}^i}\right) - \frac{\partial L}{\partial \xi^i} &= -\frac{\partial a_i}{\partial \xi^j}\dot{\xi}^j + \frac{\partial a_j}{\partial \xi^i}\dot{\xi}^j + \frac{\partial H}{\partial \xi^i} \\ &= \left(\frac{\partial a_j}{\partial \xi^i} - \frac{\partial a_i}{\partial \xi^j}\right)\dot{\xi}^j + \frac{\partial H}{\partial \xi^i} = 0\end{aligned}\quad (3.11)$$

Here we define the term $\frac{\partial a_j}{\partial \xi^i} - \frac{\partial a_i}{\partial \xi^j}$ as b_{ij} , then we have:

$$b_{ij}\dot{\xi}^j = -\frac{\partial H}{\partial \xi^i} \quad (3.12)$$

Here the $\mathbf{a}(\boldsymbol{\xi})$ is the Berry connection, while the b_{ij} is the Berry curvature.

For quantum system, here we only consider the coherent state, which means that the quantum state can be uniquely determined by a set of parameters, which is $|\psi(\boldsymbol{\xi})\rangle$. In this case the single point in phase space is exactly a quantum state. The Schrodinger equation is:

$$i\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle \quad (3.13)$$

Which is obtained from the definition of action:

$$S = \int_{t_1}^{t_2} \langle \psi | i\frac{d}{dt} - \hat{H} | \psi \rangle \quad (3.14)$$

And we can get:

$$i\langle \psi | \frac{d}{dt} |\psi\rangle - \langle \psi | \hat{H} |\psi\rangle = 0 \quad (3.15)$$

Comparing with the generalized Lagrange equation, we have:

$$i\langle \psi | \frac{d}{dt} |\psi\rangle = -a_i(\boldsymbol{\xi})\dot{\xi}^i \quad (3.16)$$

Now look into the term $i\langle \psi | \frac{d}{dt} |\psi\rangle$, which is:

$$i\langle \psi | \frac{d}{dt} |\psi\rangle = i\langle \psi | \frac{\partial}{\partial \xi^i} \dot{\xi}^i |\psi\rangle \quad (3.17)$$

$\dot{\xi}^i$ and $|\psi\rangle$ are commutable, so we have:

$$i\langle \psi | \frac{d}{dt} |\psi\rangle = i\langle \psi | \frac{\partial}{\partial \xi^i} |\psi\rangle \dot{\xi}^i \quad (3.18)$$

Thus we have:

$$a_i = -i\langle \psi | \frac{\partial}{\partial \xi^i} |\psi\rangle \quad (3.19)$$

4 Spin

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Spin is some internal property of particles, like mass. To get the value of it, we need a tool that is called spin operator $\mathbf{S}$. The spin operator is a vector and the basis of this operator space is Pauli matrix. The explicit expression for spin operator is:

$$\hat{\mathbf{S}} = \frac{\hbar}{2} \boldsymbol{\sigma} \quad (4.1)$$

The detailed expression for Pauli matrix is:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.2)$$

Note here the representation of the Pauli matrix is based on the basis $|\uparrow\rangle$ and $|\downarrow\rangle$, which means the up-spin and down-spin of electrons along the z-axis.

4.1 Rotation of spin

The expression for the rotation of quantum state is that:

$$R(\theta, \hat{\mathbf{n}}) = \exp\left(-\frac{i}{\hbar} \theta \hat{\mathbf{n}} \cdot \hat{\mathbf{J}}\right) \quad (4.3)$$

Where $\hat{\mathbf{n}}$ is the axis of the rotation, and the θ is the angle. Considering the Lie algebra of the Pauli matrix has the same structure as the rotation, so we can define the rotation of spin as:

$$D(\theta, \hat{\mathbf{n}}) = \exp\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right) \quad (4.4)$$

For the expression, we can expand it in Taylor series, which gives:

$$D(\theta, \hat{\mathbf{n}}) = I + \left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right) + \frac{\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)^2}{2!} + \frac{\left(-\frac{i}{2} \theta \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)^3}{3!} + \dots \quad (4.5)$$

It can be noted that:

$$(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})^2 = I \quad (4.6)$$

So the formulation can be rewritten as:

$$\begin{aligned} D(\theta, \hat{\mathbf{n}}) &= \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m}\right) I - i \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1}\right) (\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) \\ &= \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) (\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) \end{aligned} \quad (4.7)$$

We may observe that the existence of $\frac{\theta}{2}$, which means the rotation of spin is not SO(3) symmetry. It will not return to the original point when you rotate it as 2π . Instead, you have to rotate it to 4π as a periodic.

4.2 Expression for the eigenstates of spin in any direction

To get the eigenstates of the spin in any direction, it's intuitively that we may think that, under a rotation along the axis which has the same direction, the spin should not change. So the up-spin along that axis should have the equation:

$$\frac{1}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} |\psi\rangle = \frac{1}{2} |\psi\rangle \quad (4.8)$$

The $\hat{\mathbf{n}}$ vector can be written as: $\hat{\mathbf{n}} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$. And we have:

$$\begin{aligned} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} &= \begin{pmatrix} \cos \theta & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \end{aligned} \quad (4.9)$$

For any arbitrary states, we can denote it as $\begin{pmatrix} u \\ v \end{pmatrix}$. So we have:

$$\begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \quad (4.10)$$

And we can get a relation between u and v :

$$\begin{aligned} u \cos \theta + v \sin \theta e^{-i\phi} &= u \\ v(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}) e^{-i\phi} &= 2u \sin^2 \frac{\theta}{2} \\ \frac{u}{v} &= \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2} e^{i\phi}} \end{aligned} \quad (4.11)$$

Thus the eigenstates of the spin in any direction can be expressed as $\begin{pmatrix} e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \\ e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \end{pmatrix}$.

A trick for the calculation is that, $\langle \hat{\mathbf{n}} | \sigma_z | \hat{\mathbf{n}} \rangle = \cos \theta$, where θ is the angle between the direction vector and the z axis. Same for σ_x and σ_y , as $\langle \sigma_x \rangle = \sin \theta \cos \phi$, $\langle \sigma_y \rangle = \sin \theta \sin \phi$.

4.3 Zeeman effect

Zeeman effect is that, the magnetic moment will couple with the external magnetic field $\mathbf{B}$, which gives rise to the energy difference. The Hamiltonian for this effect is:

$$\bar{H} = -\boldsymbol{\mu} \cdot \mathbf{B} \quad (4.12)$$

The magnetic moment usually has the structure as:

$$\boldsymbol{\mu} = (\text{factor}) \times (g \text{ factor}) \times (\text{angular momentum}) \quad (4.13)$$

Some typical example is:

- Angular orbital momentum: $\boldsymbol{\mu}_L = -\frac{\mu_B}{\hbar} \mathbf{L}, g \approx 1$
- Spin magnetic momentum: $\boldsymbol{\mu}_S = -2\frac{\mu_B}{\hbar} \mathbf{S}, g \approx 2$
- Total angular momentum: $\boldsymbol{\mu}_J = -g_J \frac{\mu_B}{\hbar} \mathbf{J}$

4.4 Spin orbital coupling

In crystal configuration, it may have the expression as:

$$\bar{H} = g\sigma^i \bar{p}_i \quad (4.14)$$

Which is contributed by the relativity effect, where the electrons couple with the magnetic field generated by it self. So there will have the momentum component in the expression.

4.5 Larmor precession

Here we choose the up spin along z axis as $|\uparrow\rangle$, and the magnetic field is $\mathbf{B} = B\hat{\mathbf{z}}$, and set $\hbar = 1$. The eigenstate along $\hat{\mathbf{n}}$ can be written as:

$$|\hat{\mathbf{n}}\rangle = e^{-\frac{\phi}{2}i} \cos \frac{\theta}{2} |\uparrow\rangle + e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} |\downarrow\rangle \quad (4.15)$$

So the Lagrange of the system can be written as:

$$L = \langle \hat{\mathbf{n}} | i \frac{d}{dt} | \hat{\mathbf{n}} \rangle - \langle \hat{\mathbf{n}} | \hat{H} | \hat{\mathbf{n}} \rangle \quad (4.16)$$

The first term is:

$$\langle \hat{\mathbf{n}} | i \frac{d}{dt} | \hat{\mathbf{n}} \rangle = \frac{1}{2} \cos \theta \dot{\phi} \quad (4.17)$$

And the second term is:

$$\langle \hat{\mathbf{n}} | \hat{H} | \hat{\mathbf{n}} \rangle = \langle \hat{\mathbf{n}} | -\mathbf{B} \cdot \hat{\mathbf{S}} | \hat{\mathbf{n}} \rangle = -\frac{1}{2} B \cos \theta \quad (4.18)$$

The Lagrange is:

$$L = \frac{1}{2} \cos \theta \dot{\phi} + \frac{1}{2} B \cos \theta \quad (4.19)$$

And we can get the equation of motion:

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = -\frac{1}{2} B \sin \theta - \frac{1}{2} \dot{\phi} \sin \theta = 0 \quad (4.20)$$

$$\frac{\partial L}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{1}{2} \sin \theta \dot{\theta} = 0 \quad (4.21)$$

Which gives that:

$$\dot{\phi} = -B \quad (4.22)$$

$$\dot{\theta} = 0 \quad (4.23)$$

It's exactly $\frac{d}{dt} \hat{\mathbf{n}} = \hat{\mathbf{n}} \times \mathbf{B}$. It's so called Larmor precession.

5 Periodic crystal

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For crystal, we can always find its primitive cell. The whole crystal configuration can be treated as the translation of the primitive cell. If we assume the repeated primitive cell as infinite, then the system will have the translation symmetry, which means that, the system is unchanged under the translation operation. In the real case, it's not real infinite but we can approximately treat it as infinite. To keep the translation symmetry under finite number of repeated units, one solution is that, we can wrap it as a circle. For example, in 1D case, if we assume there are N repeated primitive cell, labeled as $0, 1, 2, \dots, N-1$, then we define the translation operator T , which function as translate the i -th primitive cell to the $i+1$. For the unit near the boundary, like $N-1$, it will be move to the 0. So it actually looks like a ring.

5.1 Bloch Basis

Since the crystal lattice is composed by the repeating of primitive cell, we can assume that, for each primitive cell, there are same localized wave functions. These wave functions are orthonormal to each other. Bloch basis is that we pack all the locally same wave functions into a periodic function. For 1D case, assume the localized wave function as $|\psi(x)\rangle$, with lattice constant as a . The Bloch basis is defined as:

$$\begin{aligned} |\Psi(x)\rangle = & \frac{1}{\sqrt{N}} [|\psi(x)\rangle + e^{ika} |\psi(x-a)\rangle + e^{2ika} |\psi(x-2a)\rangle + \dots \\ & + e^{(N-1)ika} |\psi(x-(N-1)a)\rangle] \end{aligned} \quad (5.1)$$

With $k = 0, \frac{2\pi}{Na}, \frac{4\pi}{Na}, \dots, \frac{2(N-1)\pi}{Na}$. To explain the construction of the Bloch basis, we have to explain the discrete Fourier transformation first.

Assuming we have a set of data, $x_0, x_1, \dots, x_{N-1}$. The set of data obeys the symmetry as, $x_{N+i} = x_i$. Define the translation operator T , and use $\mathbf{x}$ to denote the set, so we have:

$$T^N \mathbf{x} = \mathbf{x} \quad (5.2)$$

The eigenvalue of the operator T^N is 1. Assuming we have eigenvector $|k\rangle$:

$$T\mathbf{k} = \lambda\mathbf{k} \quad (5.3)$$

Then we have relation $\lambda^N = 1$, with solution $\lambda = e^{i\frac{2\pi k}{N}}$, with $k = 0, 1, \dots, N-1$ (Note: in physics we usually incorporate the factor $\frac{2\pi}{N}$ to represent the wave vector). We can set the orthonormal basis $|n\rangle$, with expression $\langle i|n\rangle = \delta_{ni}$. So the eigenvector $\mathbf{k}$ can be definitely expressed as:

$$|k\rangle = \sum_{n=0}^{N-1} C_n |n\rangle \quad (5.4)$$

By applying the translation operator T , we have $T|i\rangle = |i+1\rangle$, thus:

$$T|k\rangle = T \sum_{n=0}^{N-1} C_n |n\rangle = \sum_{n=1}^{N-1} C_{n-1} |n\rangle + C_{N-1} |0\rangle = e^{i\frac{2\pi k}{N}} |k\rangle \quad (5.5)$$

Thus we have:

$$\begin{aligned} \frac{C_{n-1}}{C_n} &= e^{i\frac{2\pi k}{N}}, (n \neq 0) \\ \frac{C_{N-1}}{C_0} &= e^{i\frac{2\pi k}{N}} \end{aligned} \quad (5.6)$$

A reasonable selection is that, $C_0 = \frac{1}{\sqrt{N}}$, so the expression for $|k\rangle$ is:

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{-i\frac{2\pi kn}{N}} |n\rangle \quad (5.7)$$

And we can get the discrete Fourier transformation matrix as $F_{kn} = \frac{1}{\sqrt{N}} e^{-i\frac{2\pi kn}{N}}$.

The derivation above explains the selection of Bloch basis.

5.2 1D Tight Binding Model

Assuming a 1D infinite lattice with lattice constant a , each primitive cell there is one localized orbital $|\psi(x)\rangle$. Set the hopping term $\langle\psi(x+a)|H|\psi(x)\rangle = -t$, where t is real, and the eigenvalue for the localized orbital as $\langle\psi|H|\psi\rangle = \varepsilon_0$. Using the convention above to set the Bloch basis, then we have:

$$E(k) = \langle\Psi|H|\Psi\rangle = \varepsilon_0 - 2t \cos ka \quad (5.8)$$

Which is the so called dispersion relation.

For 1D infinite lattice with lattice constant a , with each primitive cell has two localized orbital $|\psi_A(x)\rangle$ and $|\psi_B(x)\rangle$. The 1D lattice is like a chain $-A-B-A-B-A-B-$. We can define two Bloch basis:

$$\begin{aligned} |\Psi_A(x)\rangle &= \frac{1}{\sqrt{N}} [|\psi_A(x)\rangle + e^{ika} |\psi_A(x-a)\rangle + e^{2ika} |\psi_A(x-2a)\rangle + \dots \\ &\quad + e^{(N-1)ika} |\psi_A(x-(N-1)a)\rangle] \end{aligned} \quad (5.9)$$

$$\begin{aligned} |\Psi_B(x)\rangle &= \frac{1}{\sqrt{N}} [|\psi_B(x)\rangle + e^{ika} |\psi_B(x-a)\rangle + e^{2ika} |\psi_B(x-2a)\rangle + \dots \\ &\quad + e^{(N-1)ika} |\psi_B(x-(N-1)a)\rangle] \end{aligned} \quad (5.10)$$

The Hamiltonian turns into a 2×2 matrix. Set $\langle \psi_A | H | \psi_A \rangle = \varepsilon_A$, $\langle \psi_B | H | \psi_B \rangle = \varepsilon_B$. Assuming the hopping only happens between the connecting neighbors, set $\langle \psi_A | H | \psi_B \rangle = -t$. The Hamiltonian is:

$$H = \begin{pmatrix} \varepsilon_A & -t - te^{-ika} \\ -t - te^{ika} & \varepsilon_B \end{pmatrix} \quad (5.11)$$

Solving the eigenvalue for the Hamiltonian matrix, we have:

$$\begin{aligned} (E - \varepsilon_A)(E - \varepsilon_B) &= 2t^2(1 + \cos ka) \\ E^2 - (\varepsilon_A + \varepsilon_B)E + \varepsilon_A\varepsilon_B &= 2t^2(1 + \cos ka) \end{aligned} \quad (5.12)$$

Thus the dispersion relation is:

$$E(k) = \frac{(\varepsilon_A + \varepsilon_B) \pm \sqrt{(\varepsilon_A + \varepsilon_B)^2 - 4[\varepsilon_A\varepsilon_B - 2t^2(1 + \cos ka)]}}{2} \quad (5.13)$$

Which has two band.

6 Scattering

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7 Transport Property

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7.1 Boltzmann Transport Equation

Imagine there is a phase space, which is used to characterize the dynamics of one coherent state. Assuming the system is composed by a set of coherent state, with each approximately treated as nearly independent to each other. Then we have a distribution function, $g(\boldsymbol{\xi}, t)$, denotes the density of states at certain point. For Boson, the distribution obeys $g_0(\boldsymbol{\xi}) = \frac{1}{e^{\beta[H(\boldsymbol{\xi})-\mu]}-1}$. For Fermion, the distribution obeys $g_0(\boldsymbol{\xi}) = \frac{1}{e^{\beta[H(\boldsymbol{\xi})-\mu]}+1}$. Here μ is the chemical potential. For equilibrium, we have condition:

$$\frac{dg}{dt} = 0 \quad (7.1)$$

which gives:

$$\frac{\partial g}{\partial t} + \xi^i \frac{\partial g}{\partial \xi^i} = 0 \quad (7.2)$$

Replace the $\boldsymbol{\xi}$ with $\boldsymbol{x}$ and $\boldsymbol{k}$, we have:

$$\frac{\partial g}{\partial t} + x^i \frac{\partial g}{\partial x^i} + k^i \frac{\partial g}{\partial k^i} = 0 \quad (7.3)$$

For non-equilibrium state, we have equation $\frac{dg}{dt} = C[g]$, where $C[g]$ is the functional of g , stands for collision term. Under the relaxation time approximation (RTA), we have the equation:

$$C[g] \approx -\frac{g - g_0}{\tau} \quad (7.4)$$

So the Boltzmann transport equation now is:

$$\frac{\partial g}{\partial t} + x^i \frac{\partial g}{\partial x^i} + k^i \frac{\partial g}{\partial k^i} = -\frac{g - g_0}{\tau} \quad (7.5)$$

7.2 Intrinsic Anomalous Hall Effect

Under the linear response approximation, we can assume that, the density of states can be written as $g = g_0 + \delta g$, where δg is the small deviation from equilibrium. In which case we can calculate the current density at certain location:

$$\begin{aligned} \boldsymbol{J} &= \frac{-e}{(2\pi)^3} \int g(\boldsymbol{\xi}, t) \boldsymbol{v} d^3k \\ &= \frac{-e}{(2\pi)^3} \int (g_0 + \delta g) \boldsymbol{v} d^3k \end{aligned} \quad (7.6)$$

Where the factor $\frac{1}{(2\pi)^3}$ is given by the relation $\frac{1}{V} \sum_{\boldsymbol{k}} \longrightarrow \int \frac{d^3k}{(2\pi)^3}$. To solve out the explicit expression, we have to determine the form of δg . We can set the steady state assumption, which means that, $\frac{\partial g}{\partial t} = 0$. And we have equation:

$$x^i \frac{\partial g}{\partial x^i} + k^i \frac{\partial g}{\partial k^i} = -\frac{g - g_0}{\tau} \quad (7.7)$$

Using the equation $g = g_0 + \delta g$, we can get:

$$\dot{x}^i \frac{\partial}{\partial x^i} (g_0 + \delta g) + \dot{k}^i \frac{\partial}{\partial k^i} (g_0 + \delta g) = -\frac{\delta g}{\tau} \quad (7.8)$$

The general Lagrange for coherent states is:

$$L = k_i \dot{x}^i - a_i(\mathbf{x}) \dot{x}^i - \tilde{a}_i(\mathbf{k}) \dot{k}^i - H \quad (7.9)$$

Applying Lagrange equation, we have:

$$\dot{k}_i + b_{ij} \dot{x}^j = -\frac{\partial H}{\partial x^i} \quad (7.10)$$

$$-\dot{x}^i + \tilde{b}_{ij} \dot{k}^j = -\frac{\partial H}{\partial k^i} \quad (7.11)$$

For the system that we only apply an external electric field, the Hamiltonian for a single electron is:

$$H = \epsilon(\mathbf{k}) - eE_i x^i \quad (7.12)$$

Where $\epsilon(\mathbf{k})$ is the dispersion relation. The term b_{ij} is 0 due to there's no external magnetic field. Also we can get:

$$\frac{\partial H}{\partial x^i} = -eE_i \quad (7.13)$$

$$\frac{\partial H}{\partial k^i} = \frac{\partial \epsilon}{\partial k^i} \quad (7.14)$$

Thus we can get:

$$\dot{k}_i = eE_i \quad (7.15)$$

$$\dot{x}^i = \frac{\partial \epsilon}{\partial k^i} + \tilde{b}_{ij} \dot{k}^j \quad (7.16)$$

Considering linear response approximation, which means that we only keep up to the first order term. $\dot{x}^i$ is zero order term because it is the intrinsic property of the band structure. $\dot{k}^i$ is first order term because it's driven by the external electric field, which is first order term. $\frac{\partial}{\partial \xi} g_0$ is zero order term because it's intrinsic property of the distribution. So we can reduce the equation to:

$$\dot{x}^i \frac{\partial}{\partial x^i} g_0 + \dot{k}^i \frac{\partial}{\partial k^i} g_0 = -\frac{\delta g}{\tau} \quad (7.17)$$

Reorganize the equation, and we can get the expression for δg :

$$\left(\frac{\partial \epsilon}{\partial k^i} + \tilde{b}_{ij} \dot{k}^j \right) \frac{\partial}{\partial x^i} g_0 + eE_i \frac{\partial}{\partial k^i} g_0 = -\frac{\delta g}{\tau} \quad (7.18)$$

Then we have current density:

$$J^\alpha = \frac{-e}{(2\pi)^3} \int \left[-\tau \left(\frac{\partial \epsilon}{\partial k^i} + \tilde{b}_{ij} \dot{k}^j \right) \frac{\partial}{\partial x^i} g_0 - \tau e E_i \frac{\partial}{\partial k^i} g_0 + g_0 \right] \left(\frac{\partial \epsilon}{\partial k^\alpha} + \tilde{b}_{\alpha j} \dot{k}^j \right) d^3 k \quad (7.19)$$

The space is approximately treated as homogeneous, and for equilibrium distribution g_0 , there's no gradient $\frac{\partial g_0}{\partial x^i} = 0$. Then the expression is:

$$J^\alpha = \frac{-e}{(2\pi)^3} \int (-\tau e E_i \frac{\partial g_0}{\partial k^i} + g_0) (\frac{\partial \epsilon}{\partial k^\alpha} + \tilde{b}_{\alpha j} \dot{k}^j) d^3k \quad (7.20)$$

There are four terms. $g_0 \frac{\partial \epsilon}{\partial k^\alpha}$ will be 0 because it's the equilibrium case. The $\tau e E_i \frac{\partial g_0}{\partial k^i} \tilde{b}_{\alpha j} \dot{k}^j$ term will be ignored because $\dot{k}_j = e E_j$. Thus it's second order term. In the end, we get the expression:

$$\begin{aligned} J^\alpha &= \frac{1}{(2\pi)^3} \int \tau e^2 E_i \frac{\partial g_0}{\partial k^i} \frac{\partial \epsilon}{\partial k^\alpha} - e g_0 \tilde{b}_{\alpha j} \dot{k}^j d^3k \\ &= \frac{1}{(2\pi)^3} \int \tau e^2 E_i \frac{\partial g_0}{\partial k^i} \frac{\partial \epsilon}{\partial k^\alpha} - e^2 g_0 \tilde{b}_{\alpha j} E^j d^3k \end{aligned} \quad (7.21)$$

The first term is the classical Drude model term, while the second term is the anomalous term, which is contributed by the Berry curvature $\tilde{b}_{ij}$ of the phase space.